

Database Replication

Source: (Perform all activities as Root/Admin)

Step 1 : Create a Database.

Step 2: Modify config files:

Ubuntu : (As root user)

Open terminal

Open config file using nano : `sudo nano /etc/mysql/mysql.conf.d/mysqld.cnf`

Modify the following fields:

`bind-address : 0.0.0.0`

`Server-id : 1` -- uncomment it in the conf file.

Uncomment `log_bin` field as well

`binlog_do_db : <your database name >` that you created in step 1.

Save the file.

Windows:

Go to C:\ ProgramData\ MySQL \ MySQL Server 8.0 \ my.ini

Open using Notepad as an Administrator

Modify the following fields:

bind-address : 0.0.0.0

server-id : 1 -- uncomment it in the conf file.

Uncomment log_bin field as well

binlog_do_db : <your database name > that you created in step 1.

Save the file.

Step 3: Restart mysql service: (as root user)

Ubuntu : In Terminal : service mysql restart

Windows : Open cmd as Admin :

```
net stop <service name>
```

```
net start <service name>
```

You can find service name using : `sc query | findstr /I "mysql"`

Step 4: Grant replication access to slave

Open mysql :

```
sudo mysql -u root -p
```

Create user '<username>'@'%' identified with
mysql_native_password by '<password>';

Grant replication slave on *.* to '<username>'@'%';

Flush privileges;

```
show master status;
```

You will see a table, keep a note of the file name, position and database name. This information will be used in the replica side.

Replica:

Step1: Create Same Database as on the Source Server.

Step 2: Configure MySQL configuration file.

- Check for connection to source server and vice versa (very crucial step)
 - o ping <source ip>

If it does not work then try turning off firewall

Ubuntu: sudo ufw disable

Windows: netsh advfirewall set privateprofile state off

Ubuntu: (as root user)

- Change mysql configuration file
 - o sudo nano /etc/mysql/mysql.conf.d/mysqld.cnf

Edit the details given below (don't forget to uncomment) :

```
server-id = 2 #change server id to 2
log_bin = /var/log/mysql/mysql-bin.log
binlog_expire_logs_seconds = 2592000
max_binlog_size = 100M
```

```
binlog_do_db = mydatabase #database name created  
earlier as on the source  
relay_log = /var/log/mysql/mysql-relay-bin.log
```

Windows:

Go to C:\ProgramData\MySQL\MySQL Server 8.0\my.ini
Open using Notepad as an Administrator

```
#Add Details given below  
server-id = 2  
binlog_do_db = mydatabase  
relay_log = relay-bin-logs
```

Step 3: Restart mysql service: (as root user)

Ubuntu : In Terminal : service mysql restart

Windows : Open cmd as Admin :

```
net stop <service name>
```

```
net start <service name>
```

Step 4: Now in terminal open MySQL command line client

- o `sudo mysql -u root -p`
- In MySQL command line client execute given commands:
 - o `stop replica;`
 - o `Reset replica all;`
 - o `CHANGE REPLICATION SOURCE TO`
`SOURCE_HOST='<source server ip>'`,
`SOURCE_USER='<source server user>'`,
`SOURCE_PASSWORD=' <source server password> '`,
`SOURCE_LOG_FILE=' <source server log file> '`,
`SOURCE_LOG_POS= <source server POS number>;`
- `Start replica;`
- `Show replica status; -- To see if replica is connected`
-

After successful completion of connection with source server and replica creation you will see

“Waiting for source to send event” in replica status.

Now any changes made in source server it will show in Replica.